Labs 6: Natural Selection on the plant *Brassica rapa*, Part II

Reminder of what we are doing:

Does cold affect fitness in two *Brassica* phenotypes?

Purple stem with trichomes

Green stem without trichomes

This week:

1) Repeat last week's anthocyanin measurements
2) Estimate fitness of *Brassica*.
3) Do a statistical analysis on the data
4) Graph fitness data

How to collect data. Part 1. Getting tissue.

Detailed instructions are on Canvas. Read these before you do anything!!!!

Your group will be assigned a letter (A, B, C, D, E or F). This is critical to make sure you know where to place your samples in the tray for the plate reader. Labeling is a very, very important part of the scientific process.

Control Data:

-Select 4 vials: Label these with your assigned group letter.
-Pick 2 green and 2 purple plants from the tray labeled CONTROL.
-Place a toothpick at the base of each plant. This lets others know the plant was already sampled
-Remove about half a leaf from each plant
    -You should have 2 green leaves and 2 purple leaves total
-We want 0.05-0.08 g of tissue. Weigh each leaf and cut off leaf tissue until you have the correct weight.
-Cut the leaves into smaller pieces and place in the appropriate vials
-Record the weight of each leaf and the vial you placed each leaf in (very important)

Cold Treatment Data Collection

-Repeat this process for the cold treatment plants

Overview of measuring anthocyanin concentrations. Gloves and goggles required.

Be sure to follow the detailed instructions on Canvas and the instructions on what to do with waste

We will measure the concentration of anthocyanin using a plate reader

The plate reader shines visible light through the sample and measures the amount of the light the sample absorbs

This provides an absorbance value we will convert to micrograms (μg) of anthocyanin/g leaf tissue

To prepare the samples:

1) add 1mL of acidified ethanol to each vial using a pipette. This stabilizes the anthocyanin
2) Place your vials in the sonicator for 10 minutes. This helps break up the cells, releasing anthocyanin
3) Centrifuge your vials for 1 minute at 1000 rpm (be sure it is balanced)
   This pushes the leaf tissue to the bottom of the vial—we don't want that stuff
4) Using a pipette, add 200μL to the plate in the row for your group letter (Group A, place your samples in row A.
5) Plates are placed in the plate reader, and these data will be sent to you.
9) Enter your data into the class spreadsheet
10) Using class your lab's data: calculate mean μg anthocyanin/g leaf tissue, SD,
11) Graph the lab's data: mean and SD

Estimating fitness in Brassica.

Things to think about:

What is the relationship of flowers on a plant to a plant's reproductive success?
Would more flowers or fewer flowers on a plant suggest higher or lower fitness?
Why are we counting flowers rather than counting number of offspring?

Data Collection

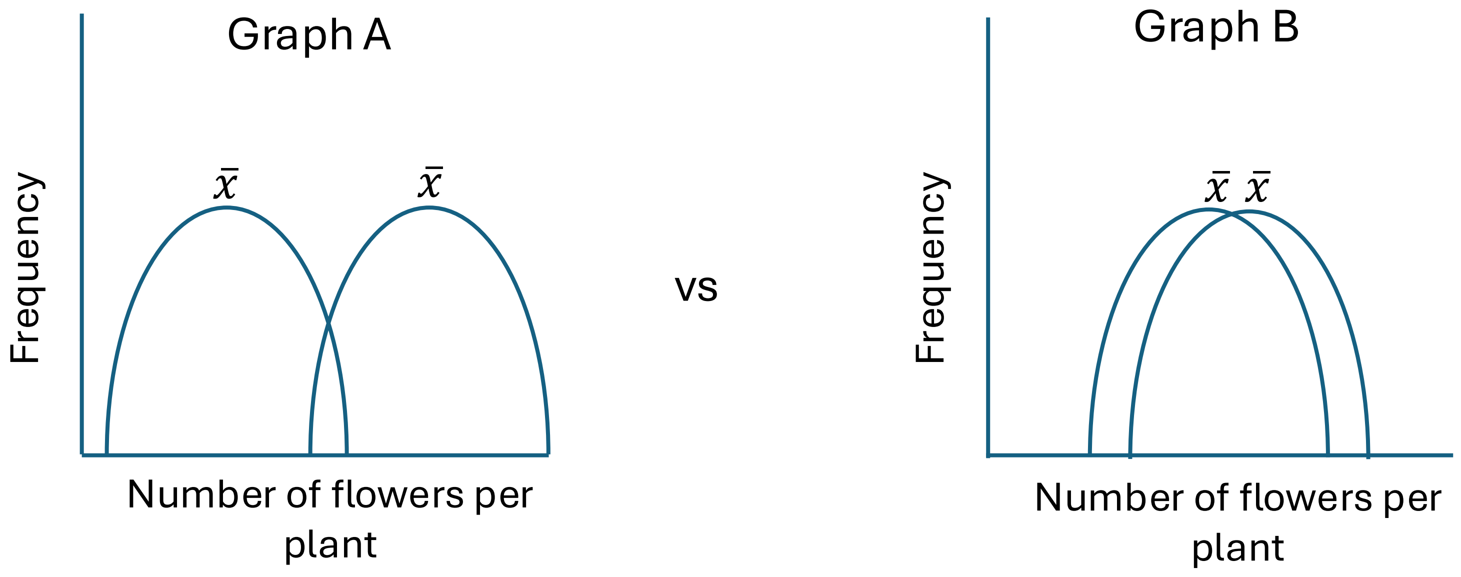
Step 1. As a class, count the total number of flowers and flower buds for each phenotype and treatment
Step 2. Record those numbers in the class spreadsheet
Step 3. Calculate the average number of flowers and SD for the green and purple plants for all treatments
Step 4. Graph the average number of flowers, and include SD error bars
Step 5. Do a statistical analysis on the flower data. Follow the instructions in the lab manual on how to do a t-test

Analyzing data using statistics

-How do we decide if the averages are truly different (significantly different) from each other?

In other words, do our data look more like Graph A or more like Graph B?

$\bar{x}$ = average value for the trait



For which data set we can be reasonably confident that the averages are different?
The t-test will help us make this determination.

t-test: A statistical test that helps us determine if two averages are different

$$t = \frac{\bar{x}_{Green} - \bar{x}_{purple}}{\sqrt{\frac{SD^2{}_{Green}}{n_{Green}} + \frac{SD^2_{purple}}{n_{purple}}}}$$

The t-test takes into account the following:

-Differences in the averages ($\bar{x}_{Green} - \bar{x}_{purple}$)
-How variable the data are (Standard deviation, SD)
-Our sample size, n. Or how many data points you have. The more data points, the more information we have

In Excel, the t-test function calculates a p-value, which tells us the likelihood the averages are different, or not

$p < 0.05$ tells us there is a significant difference; if $p > 0.05$, we can't say there is a difference.

Anthocyanin: Post Treatment Data only (this week's data)

Calculate mean for SD for each treatment

Post-Treatment Summary Data (Week 6 Lab)

	Genotype	Control	Cold Treatment
Mean	ANL/ANL Purple	0.4440	0.6091
	anl/anl Green	0.5211	0.4065
St Dev	ANL/ANL Purple	0.3052	0.2769
	anl/anl Green	0.3107	0.3152

Conduct a t-test for the following comparisons:

Do purple stem plants and green stem plants produce different quantities of anthocyanin under control conditions?

Do purple stem plants and green stem plants produce different quantities of anthocyanin under cold conditions?

T Test for comparing data across Genotypes

	Control	Cold
p value	0.95	0.03

T Test for comparing data across treatements

	ANL/ANL purple	anl/anl green
p value	0.02	0.60

Do purple stem plants in control conditions produce different quantities of anthocyanin than in cold conditions?

Do green stem plants in control conditions produce different quantities of anthocyanin than in cold conditions?

t-test example with purple control vs green control

A	B	C	D	E	F
				Anthocyanin (µg/g)	
Group	Treatment	Genotype	Replicate	Pre	Post
A	Control	Purple	1	0.5350	0.8711
A	Control	Purple	2	0.2604	0.7659
B	Control	Purple	1	0.1215	0.2524
B	Control	Purple	2	0.0927	0.1939
C	Control	Purple	1	0.2670	0.5310
C	Control	Purple	2	0.0326	0.4189
D	Control	Purple	1	0.9261	0.3378
D	Control	Purple	2	0.0959	0.9897
E	Control	Purple	1	0.9903	0.1658
E	Control	Purple	2	0.4910	0.6486
F	Control	Purple	1	0.6790	0.1149
F	Control	Purple	2	0.2442	0.3055
A	Control	Green	1	0.1280	0.1518
A	Control	Green	2	0.5861	0.1116
B	Control	Green	1	0.7579	0.8356
B	Control	Green	2	0.8959	0.4565
C	Control	Green	1	0.9069	0.9240
C	Control	Green	2	0.0629	0.2240
D	Control	Green	1	0.5133	0.8856
D	Control	Green	2	0.7887	0.0793
E	Control	Green	1	0.0438	0.7232
E	Control	Green	2	0.8110	0.1978
F	Control	Green	1	0.4602	0.7042
F	Control	Green	2	0.5702	0.3883
A	Cold	Purple	1	0.7874	0.3465

Pre-Treatment Summary Data (Week 5 Lab)

	Genotype	Control	Cold Treatment
Mean	ANL/ANL Purple	0.4069	0.5386
	anl/anl Green	0.5802	0.4244
St Dev	ANL/ANL Purple	0.3222	0.3118
	anl/anl Green	0.3263	0.3365

Post-Treatment Summary Data (Week 6 Lab)

	Genotype	Control	Cold Treatment
Mean	ANL/ANL Purple	0.4440	0.6091
	anl/anl Green	0.5211	0.4065
St Dev	ANL/ANL Purple	0.3052	0.2769
	anl/anl Green	0.3107	0.3152

T Test for comparing data across Genotypes

	Control	Cold
p value	=T.TEST(F3:F14,F15:F26,2,3)	

T Test for comparing data across treatements

	ANL/ANL purple	anl/anl green
p value		

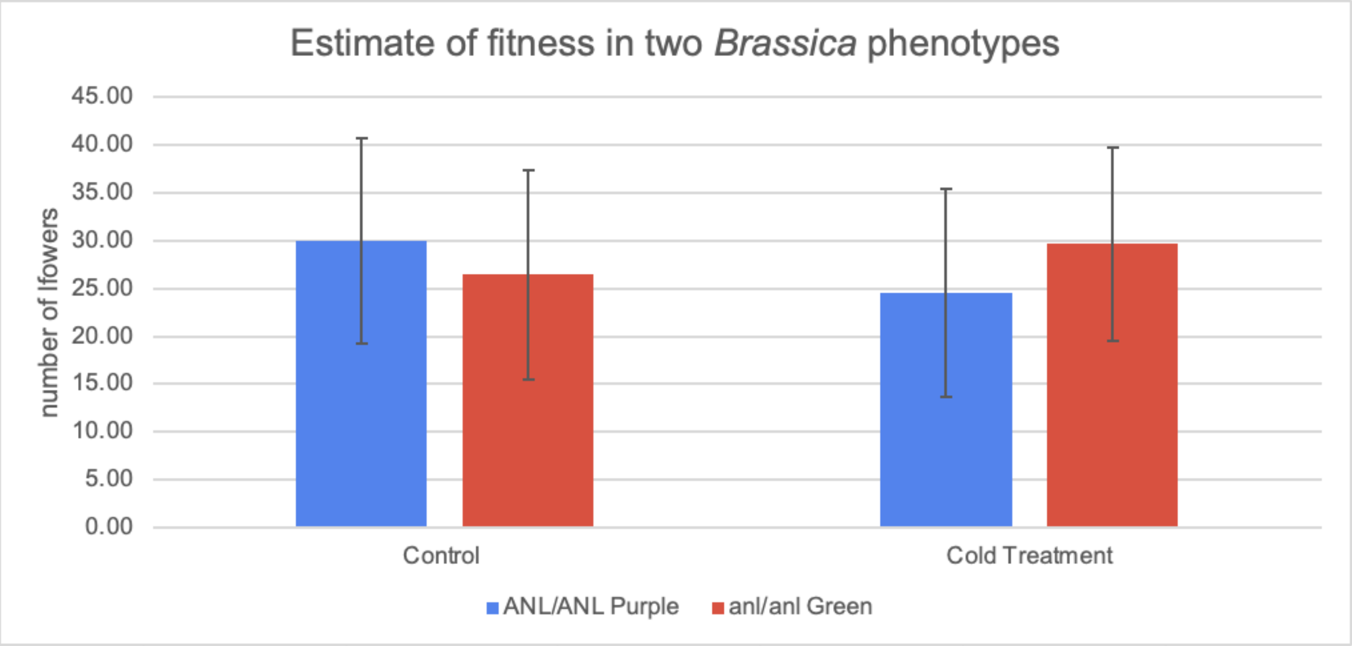
Do the same for the following comparisons:

- cold purple vs cold green
- control purple vs cold purple
- control green vs cold green

Upload a screenshot of post-treatment summary data table and p-value results of the t-tests table

Analysis of Fitness: Graph your flower data

	Genotype	Control	Cold Treatment
Mean	ANL/ANL Purple	29.97	24.53
	anl/anl Green	26.47	29.63
St Dev	ANL/ANL Purple	10.72	10.89
	anl/anl Green	10.96	10.09



Analysis of Fitness: t-test

Do purple stem plants and green stem plants produce different numbers of flowers under control conditions?

Do purple stem plants and green stem plants produce different numbers of flowers under cold conditions?

T Test for comparing data across Genotypes

	Control	Cold
p value	0.2161106085	0.06495985148

T Test for comparing data across treatements

	ANL/ANL purple	anl/anl green
p value	0.05633553318	0.2490566952

Do purple stem plants in control conditions produce different numbers of flowers than in cold conditions?

Do green stem plants in control conditions produce different numbers of flowers than in cold conditions?

t-test comparing number of flowers on green stemmed plants vs purple stemmed plants for the controls.

1	CONTROL TREATMENT				COLD TREATMENT			
2	Tray	Plant Genotype	Plant ID	Flowers	Tray	Plant Genotype	Plant ID	Flowers
4	1	ANL/ANL Purple	2	28	1	ANL/ANL Purple	2	38
5	1	ANL/ANL Purple	3	13	1	ANL/ANL Purple	3	17
6	1	ANL/ANL Purple	4	44	1	ANL/ANL Purple	4	29
7	1	ANL/ANL Purple	5	42	1	ANL/ANL Purple	5	27
8	1	ANL/ANL Purple	6	36	1	ANL/ANL Purple	6	17
9	1	ANL/ANL Purple	7	23	1	ANL/ANL Purple	7	14
10	1	ANL/ANL Purple	8	43	1	ANL/ANL Purple	8	19
11	1	ANL/ANL Purple	9	47	1	ANL/ANL Purple	9	17
12	1	ANL/ANL Purple	10	44	1	ANL/ANL Purple	10	12
13	1	ANL/ANL Purple	11	27	1	ANL/ANL Purple	11	35
14	1	ANL/ANL Purple	12	40	1	ANL/ANL Purple	12	17
15	1	ANL/ANL Purple	13	22	1	ANL/ANL Purple	13	37
16	1	ANL/ANL Purple	14	17	1	ANL/ANL Purple	14	43
17	1	ANL/ANL Purple	15	16	1	ANL/ANL Purple	15	12
18	2	ANL/ANL Purple	1	27	2	ANL/ANL Purple	1	14
19	2	ANL/ANL Purple	2	34	2	ANL/ANL Purple	2	31
20	2	ANL/ANL Purple	3	16	2	ANL/ANL Purple	3	33
21	2	ANL/ANL Purple	4	19	2	ANL/ANL Purple	4	15
22	2	ANL/ANL Purple	5	48	2	ANL/ANL Purple	5	19
23	2	ANL/ANL Purple	6	27	2	ANL/ANL Purple	6	12
24	2	ANL/ANL Purple	7	21	2	ANL/ANL Purple	7	14
25	2	ANL/ANL Purple	8	37	2	ANL/ANL Purple	8	39
26	2	ANL/ANL Purple	9	18	2	ANL/ANL Purple	9	24
27	2	ANL/ANL Purple	10	19	2	ANL/ANL Purple	10	33
28	2	ANL/ANL Purple	11	32	2	ANL/ANL Purple	11	20
29	2	ANL/ANL Purple	12	40	2	ANL/ANL Purple	12	15
30	2	ANL/ANL Purple	13	28	2	ANL/ANL Purple	13	14
31	2	ANL/ANL Purple	14	19	2	ANL/ANL Purple	14	30
32	2	ANL/ANL Purple	15	31	2	ANL/ANL Purple	15	44
33	1	anl/anl Green	1	19	1	anl/anl Green	1	31
34	1	anl/anl Green	2	15	1	anl/anl Green	2	34
35	1	anl/anl Green	3	14	1	anl/anl Green	3	24
36	1	anl/anl Green	4	23	1	anl/anl Green	4	28
37	1	anl/anl Green	5	15	1	anl/anl Green	5	25
38	1	anl/anl Green	6	44	1	anl/anl Green	6	40
39	1	anl/anl Green	7	46	1	anl/anl Green	7	39
40	1	anl/anl Green	8	22	1	anl/anl Green	8	23
41	1	anl/anl Green	9	35	1	anl/anl Green	9	26
42	1	anl/anl Green	10	39	1	anl/anl Green	10	44
43	1	anl/anl Green	11	28	1	anl/anl Green	11	43
44	1	anl/anl Green	12	22	1	anl/anl Green	12	12
45	1	anl/anl Green	13	40	1	anl/anl Green	13	35
46	1	anl/anl Green	14	14	1	anl/anl Green	14	32
47	1	anl/anl Green	15	39	1	anl/anl Green	15	40
48	2	anl/anl Green	1	18	2	anl/anl Green	1	30
49	2	anl/anl Green	2	21	2	anl/anl Green	2	42
50	2	anl/anl Green	3	19	2	anl/anl Green	3	17
51	2	anl/anl Green	4	32	2	anl/anl Green	4	16
52	2	anl/anl Green	5	16	2	anl/anl Green	5	39
53	2	anl/anl Green	6	44	2	anl/anl Green	6	40
54	2	anl/anl Green	7	33	2	anl/anl Green	7	14
55	2	anl/anl Green	8	12	2	anl/anl Green	8	26
56	2	anl/anl Green	9	13	2	anl/anl Green	9	21
57	2	anl/anl Green	10	34	2	anl/anl Green	10	40
58	2	anl/anl Green	11	27	2	anl/anl Green	11	14
59	2	anl/anl Green	12	26	2	anl/anl Green	12	38
60	2	anl/anl Green	13	18	2	anl/anl Green	13	19
61	2	anl/anl Green	14	21	2	anl/anl Green	14	18
62	2	anl/anl Green	15	45	2	anl/anl Green	15	39

J	K	L	M	N	C

	Genotype	Control	Cold Treatment
Mean	ANL/ANL Purple	29.97	24.53
	anl/anl Green	26.47	29.63
St Dev	ANL/ANL Purple	10.72	10.89
	anl/anl Green	10.96	10.09

T Test for comparing data between Genotypes

	Control	Cold
=T.TEST(D3:D32,D33:D62,2,3)		

T.TEST(array1, array2, tails, type)

T Test for comparing data between treatements

	ANL/ANL purple	anl/anl green
p value	0.06	0.25